

Project Report

Design and Simulation of a 1.45 V Bandgap Reference Circuit

Analog Integrated Electronics Course Project

Item	Information
Student Name	Ahmed Lutfi Inam
Course	Analog Integrated Electronics
Software	Cadence Virtuoso
Technology/PDK	gpd090
Target VREF	1.4 V to 1.5 V
Target Temperature Coefficient	10 ppm/°C to 17 ppm/°C from -40°C to 125°C

Summary: This report explains the theory, theoretical calculation, Cadence implementation, value selection, and simulation result of a bandgap reference circuit designed to produce approximately 1.45 V with a low temperature coefficient.

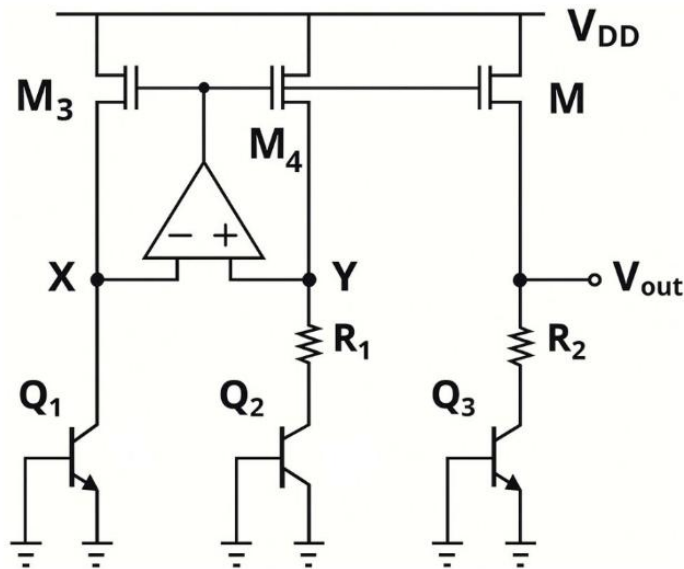
1. Project Objective and Required Specifications

The objective of this project is to design a bandgap reference voltage circuit in Cadence Virtuoso. The circuit must generate a nearly constant voltage even when temperature changes. The final required output voltage is between 1.4 V and 1.5 V, and the required temperature coefficient is between 10 ppm/°C and 17 ppm/°C over -40°C to 125°C.

Parameter	Required Value	Unit
Reference voltage, VREF	1.4 to 1.5	V
Temperature range	-40 to 125	°C
Temperature coefficient	10 to 17	ppm/°C

The baseline idea and the main equations were taken from the laboratory experiment titled "Design and Characterization of a Bandgap Reference Voltage Circuit". That lab circuit uses a PMOS current mirror, PNP bipolar transistors, resistors, and an ideal op-amp made from a voltage-controlled voltage source (VCVS).

Figure 1: Baseline bandgap reference circuit from the lab experiment



2. Bandgap Reference Theory

A bandgap reference is a circuit that generates a stable DC voltage. The main idea is to add two voltages that change in opposite directions with temperature. If these two changes are balanced properly, the final voltage becomes almost constant.

2.1 CTAT voltage: VBE

The base-emitter voltage of a BJT, V_{BE} , decreases when temperature increases. This type of voltage is called CTAT, which means Complementary To Absolute Temperature. In simple words, CTAT means the voltage goes down as temperature goes up.

V_{BE} decreases as temperature increases

For a silicon BJT, the slope of V_{BE} is usually around $-1 \text{ mV}/^{\circ}\text{C}$ to $-2 \text{ mV}/^{\circ}\text{C}$, depending on the current density and the process model.

2.2 PTAT voltage: Delta VBE

If two BJTs operate with different emitter areas or different current densities, their V_{BE} values will be slightly different. This difference is called Delta V_{BE} . Unlike V_{BE} , Delta V_{BE} increases with temperature. This type of voltage is called PTAT, which means Proportional To Absolute Temperature.

$$\Delta V_{BE} = V_T \ln(n)$$

Here, V_T is the thermal voltage and n is the effective BJT area ratio. The thermal voltage is:

$$V_T = kT/q$$

At room temperature, V_T is approximately 25.85 mV . If $n = 10$, then:

$$\Delta V_{BE} = 25.85 \text{ mV} \times \ln(10) = 59.5 \text{ mV}$$

2.3 Main VREF equation

The lab experiment gives the reference voltage equation in the following form:

$$V_{REF} = |V_{BE3}| + (R_2/R_1) V_T \ln(n)$$

This equation is the most important equation of the project. It shows that the final core reference voltage is made of two parts:

- $|V_{BE3}|$ is the CTAT part. It decreases with temperature.
- $(R_2/R_1) V_T \ln(n)$ is the PTAT part. It increases with temperature.

By choosing the resistor ratio R_2/R_1 properly, the negative slope of V_{BE} can be cancelled by the positive slope of the PTAT term. When the cancellation is good, the output voltage changes only slightly over temperature.

Symbol	Meaning in this project
V_{REF}	Final reference voltage required by the project
V_{BE3}	Base-emitter voltage of the output BJT branch
R_1	Resistor that converts Delta V_{BE} into PTAT current
R_2	Resistor that multiplies the PTAT voltage contribution
V_T	Thermal voltage, equal to kT/q
n	Effective BJT area ratio between the PTAT-generating transistors

3. Circuit Operation in Simple Words

The circuit can be divided into four important parts: the PMOS current mirror, the BJT branches, the resistor network, and the ideal op-amp.

3.1 PMOS current mirror

The three PMOS transistors at the top act like current sources. They copy current from one branch to the other branches. For better matching, all three PMOS transistors were kept equal in size.

3.2 BJT branches and multipliers

The circuit uses PNP BJTs from the gpdk090 bipolar model. The BJT multiplier means how many identical BJT unit devices are connected in parallel. For example, multiplier $m = 10$ means the device has ten unit emitter areas.

BJT branch	Multiplier used	Reason
Left/reference BJT	$m = 1$	One-unit device used as the reference current-density branch.
Middle/PTAT BJT	$m = 10$	Larger area creates a different current density and produces $\Delta V_{BE} = V_T \ln(10)$.
Right/output BJT	$m = 1$	One-unit output BJT gives higher current density and a suitable V_{BE} for the core output.

The left and middle BJT branches generate ΔV_{BE} . This ΔV_{BE} appears across R_1 and creates a PTAT current. That current then flows in the output branch through R_2 and adds a PTAT voltage to the output BJT V_{BE} .

3.3 Ideal op-amp using VCVS

The ideal op-amp is made using a voltage-controlled voltage source, VCVS, with high gain. The gain was set to 100000. This high gain forces the two input nodes of the op-amp to become almost equal. This is necessary for generating the correct ΔV_{BE} and PTAT current.

3.4 Output scaling stage

The low-TC bandgap core naturally produced about 1.17 V. This value had a good temperature coefficient but did not satisfy the required 1.4 V to 1.5 V range. Therefore, a non-inverting gain stage was added using another ideal VCVS op-amp and two resistors. This stage scales the low-TC voltage to about 1.45 V.

$$V_{REF_FINAL} = V_{out} \times (1 + R_f/R_g)$$

4. Theoretical Calculation

4.1 Choosing the BJT area ratio

The middle BJT multiplier was chosen as 10 and the left BJT multiplier was chosen as 1. Therefore, the effective area ratio was selected as:

$$n = 10$$

At room temperature:

$$V_T \approx 25.85 \text{ mV}$$

Therefore:

$$\Delta V_{BE} = V_T \ln(10) = 25.85 \text{ mV} \times 2.3026 = 59.5 \text{ mV}$$

4.2 Choosing R1 and R2 for the core

The resistor R1 controls the PTAT current. A smaller R1 gives larger current. The value R1 = 1 kOhm was used because it gives enough current for the BJT VBE branch and keeps the circuit close to the lab experiment style.

$$I_{PTAT} \approx \Delta V_{BE} / R_1 = 59.5 \text{ mV} / 1 \text{ kOhm} = 59.5 \text{ uA}$$

The resistor R2 controls how much PTAT voltage is added to the output. The ratio R2/R1 controls the temperature slope. A larger R2 increases the positive temperature slope. A smaller R2 reduces the positive temperature slope.

From theory, the best R2/R1 ratio is located by balancing the negative VBE slope with the positive PTAT slope. The simplified condition is:

$$dV_{REF}/dT \approx dV_{BE}/dT + (R_2/R_1)(k/q)\ln(n)$$

In the actual gpd090 simulation, trial-and-error and fine tuning showed that R2 = 6.85 kOhm with R1 = 1 kOhm gave the best low-TC region.

$$R_2/R_1 = 6.85 \text{ kOhm} / 1 \text{ kOhm} = 6.85$$

This value gave a core voltage near 1.17 V with only a few millivolts of variation over -40°C to 125°C.

4.3 Calculating the output scaling gain

The low-TC core output was approximately:

$$V_{out} \approx 1.17 \text{ V}$$

The target final output was selected near the center of the required range:

$$V_{REF_FINAL} \approx 1.45 \text{ V}$$

Required voltage gain:

$$A = 1.45 / 1.17 = 1.24$$

A non-inverting amplifier has the gain:

$$A = 1 + R_f/R_g$$

By choosing Rg = 100 kOhm and Rf = 24 kOhm:

$$A = 1 + 24k/100k = 1.24$$

So the final expected output becomes:

$$V_{REF_FINAL} = 1.17 \text{ V} \times 1.24 = 1.45 \text{ V}$$

5. Final Component Values and Justification

Component/value	Chosen value	Why this value was used
VDC	3.3 V	Provides enough headroom for 1.45 V output, PMOS current mirror, BJTs, and op-amp. It is also the supply used in the baseline lab setup.
R1	1 kOhm	Sets the PTAT current. With $n = 10$, it gives about 59.5 μA , which is high enough for a practical VBE level.
R2	6.85 kOhm	This value was found to be the best low-TC region after theoretical estimation and trial-and-error tuning.
BJT multipliers	1 : 10 : 1	The 1:10 pair creates Delta VBE. The output BJT was kept $m = 1$ for suitable VBE and simple matching.
PMOS W	2.4 μm	The lab used $W/L = 480\text{n}/100\text{n} = 4.8$. With $L = 500\text{n}$, $W = 2.4 \mu\text{m}$ keeps the same W/L ratio.
PMOS L	500 nm	Longer than minimum length to improve analog current mirror behavior and reduce channel-length modulation effects.
PMOS multiplier	1	All PMOS devices are equal current mirror devices. Keeping $m = 1$ makes matching simple.
VCVS gain	100000	Used as an ideal op-amp so that the input nodes are forced nearly equal.
Rg	100 kOhm	Moderate value that does not heavily load the core output.
Rf	24 kOhm	Together with Rg, gives a gain of 1.24 to scale 1.17 V to about 1.45 V.

5.1 Why trial-and-error was still needed

The equations give the design direction, but the actual Cadence result depends on real device models. The BJT VBE slope, resistor behavior, PMOS current mirror accuracy, op-amp polarity, and operating point are not perfectly ideal. Therefore, R2 was adjusted in simulation to find the region where the temperature curve became nearly flat.

Trial value	Observed behavior	Decision
R2 = 10.2 kOhm	Output moved toward 1.4 V, but temperature slope was too positive.	Rejected for high TC.
R2 = 8.2 kOhm	Slope reduced, but still too positive and TC was still high.	Rejected for high TC.
R2 = 6.85 kOhm	Core output became nearly flat over temperature, around 1.17 V.	Accepted as low-TC core.
Gain stage 1.24x	Scaled low-TC core to about 1.45 V.	Accepted as final output solution.

6. Cadence Implementation Steps

The following steps summarize how the final project was implemented in Cadence Virtuoso.

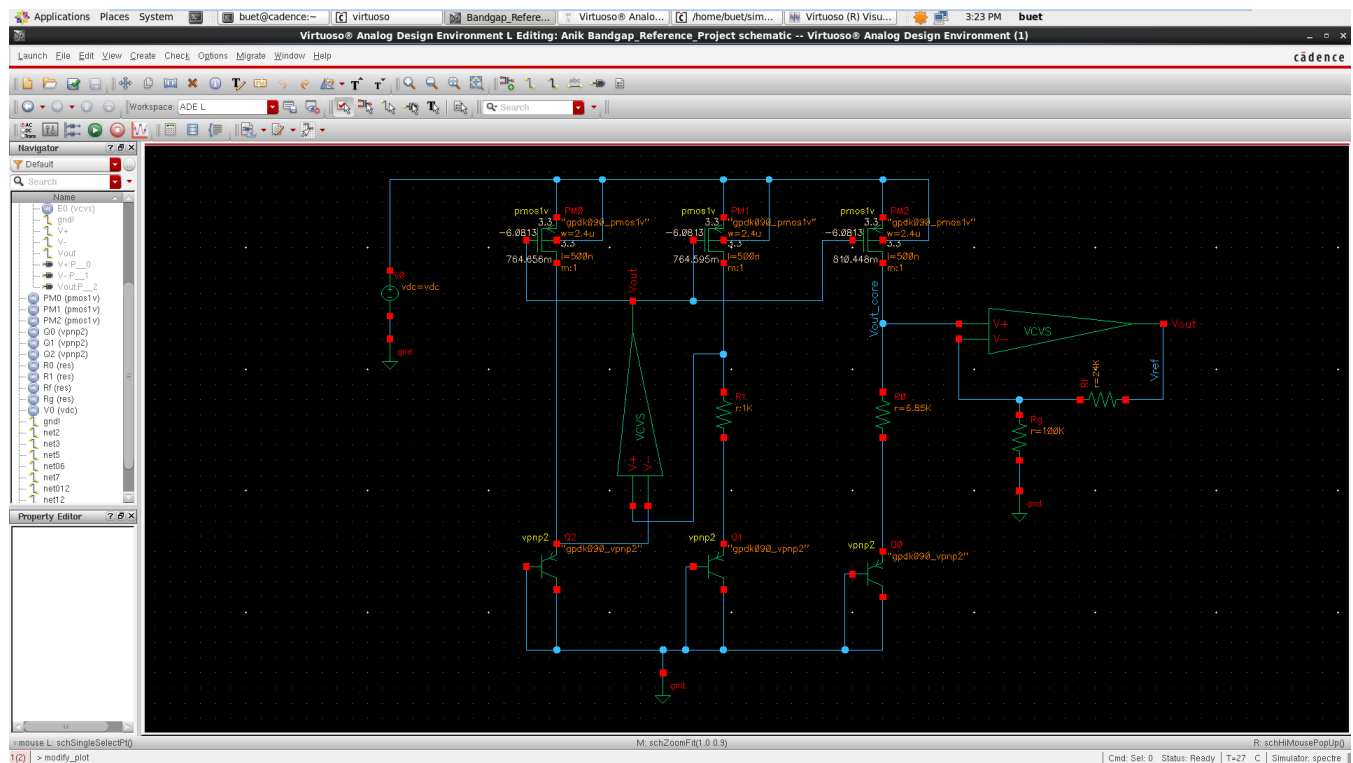
6.1 Bandgap core setup

- Create or copy the baseline bandgap reference schematic from the lab experiment.
- Use PMOS transistors from gpdk090 and PNP BJTs using the gpdk090 bipolar model.
- Set all three PMOS devices to $W = 2.4 \mu\text{m}$, $L = 500 \text{ nm}$, multiplier = 1.
- Set BJT multipliers as left = 1, middle = 10, right = 1.
- Set $R1 = 1 \text{ k}\Omega$ and $R2 = 6.85 \text{ k}\Omega$.
- Use a VCVS as the ideal op-amp and set its gain to 100000.
- Name the core output node as Vout.

6.2 Output scaling stage setup

- Add one additional ideal VCVS op-amp.
- Connect Vout to the positive input of the new op-amp.
- Connect $R_g = 100 \text{ k}\Omega$ from the negative input to ground.
- Connect $R_f = 24 \text{ k}\Omega$ from the final output node back to the negative input.
- Name the final output node VREF_FINAL.
- Plot both Vout and VREF_FINAL during simulation.

Figure 3: Cadence schematic of the final circuit



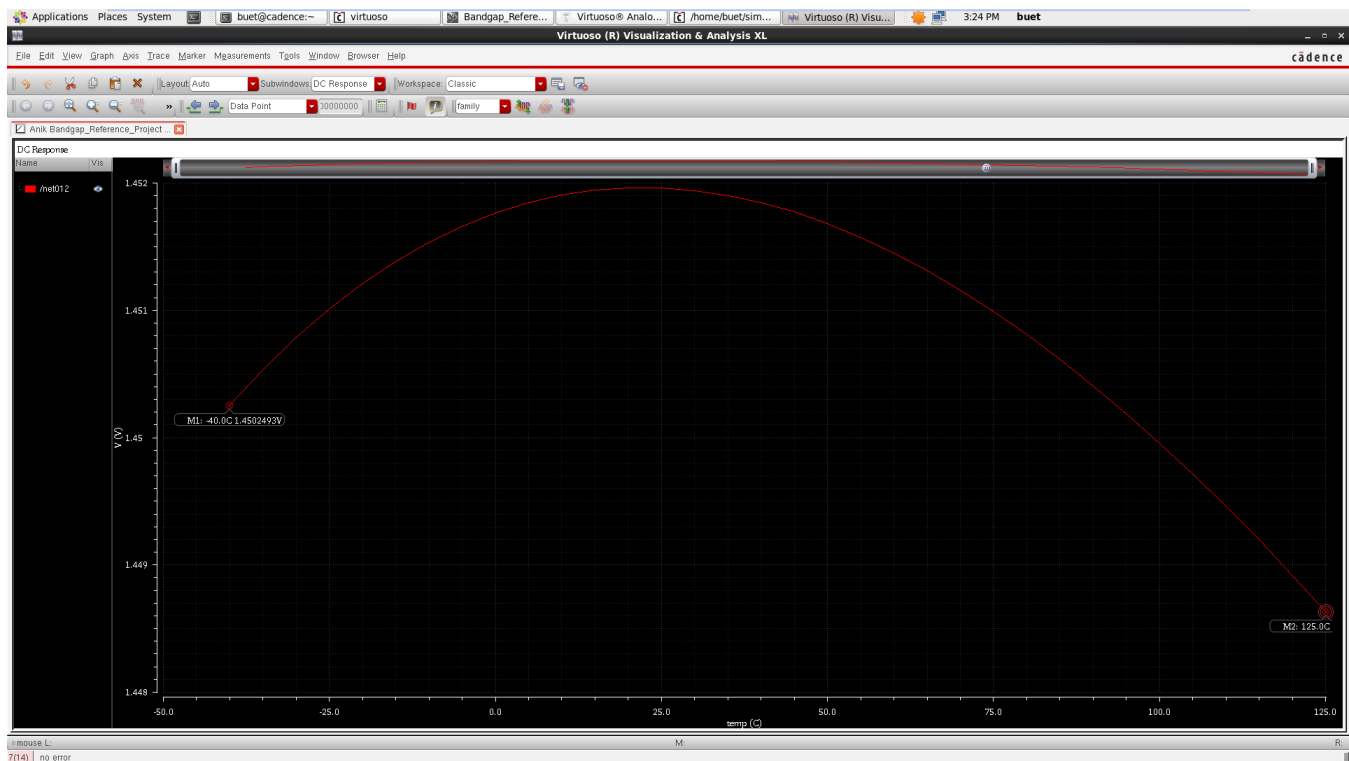
6.3 Simulation setup

Simulation item	Value used
Supply voltage, VDC	3.3 V
Nominal temperature	27°C
Temperature sweep	-40°C to 125°C
Temperature step	5°C or suitable step
Output plotted	VREF_FINAL

7. Simulation Results and Analysis

The final plotted output was VREF_FINAL. The graph shows a slightly curved shape. This is normal for a first-order bandgap reference because VBE is not perfectly linear with temperature. The important point is not whether the curve is perfectly straight. The important point is the total voltage variation and the calculated temperature coefficient.

Figure 4: Final VREF_FINAL versus temperature graph (-40°C to 125°C)



7.1 Voltage range check

From the final graph, the output stayed approximately within the following range:

Quantity	Approximate value from graph
Minimum VREF_FINAL	1.4486 V
Maximum VREF_FINAL	1.4520 V
Required range	1.4 V to 1.5 V

Therefore, the final output voltage satisfies the required VREF range.

7.2 Temperature coefficient calculation

The temperature coefficient was calculated using the max-min method:

$$TC = [(V_{max} - V_{min}) / (V_{nom} \times \Delta T)] \times 10^6$$

Using the final graph:

$$V_{max} \approx 1.4520 \text{ V}$$

$$V_{min} \approx 1.4486 \text{ V}$$

$$V_{nom} \approx 1.45 \text{ V}$$

$$\Delta T = 125 - (-40) = 165^\circ\text{C}$$

Therefore:

$$TC = [(1.4520 - 1.4486) / (1.45 \times 165)] \times 10^6$$

$$TC \approx 14.2 \text{ ppm}/^\circ\text{C}$$

The required temperature coefficient is 10 ppm/°C to 17 ppm/°C. The calculated value is approximately 14.2 ppm/°C, so the design satisfies the TC specification.

7.3 Why the graph has a peak

The final graph is slightly parabolic. It first increases slightly and then decreases slightly. This means the slope is positive at lower temperature, close to zero near the peak, and negative at higher temperature. This is not a problem. It means the positive PTAT slope and the negative CTAT slope cancel near the middle of the temperature range. Since the total variation is only a few millivolts, the design is acceptable.

8. Discussion

The original lab circuit was useful as a baseline because it already contained the essential bandgap reference structure. However, the lab circuit was not directly suitable for the project specification. The project needed a higher output voltage and a wider temperature range. Therefore, the existing circuit was redesigned and tuned.

At first, attempts were made to directly force the bandgap core output into the 1.4 V to 1.5 V range by increasing R2. This increased the voltage, but it also increased the positive temperature slope and made the TC too high. Then R2 was reduced to find the low-TC point. With R2 = 6.85 kOhm, the core output became almost temperature independent, but its level was around 1.17 V. This confirmed that the core should be optimized for TC first, and the voltage level should be adjusted separately.

The final solution uses a low-TC bandgap core followed by a non-inverting gain stage. This is a practical design method because the gain stage changes the voltage level without significantly changing the ppm/°C value, as long as the gain is temperature independent in the ideal simulation.

9. Final Design Summary

Block	Final values
Bandgap core supply	VDC = 3.3 V
Bandgap core resistors	R1 = 1 kOhm, R2 = 6.85 kOhm
BJT multipliers	Left = 1, Middle = 10, Right = 1
PMOS devices	W = 2.4 um, L = 500 nm, m = 1
Ideal op-amp gain	100000
Gain stage resistors	Rg = 100 kOhm, Rf = 24 kOhm
Final VREF range	Approximately 1.4486 V to 1.4520 V
Final TC	Approximately 14.2 ppm/°C

10. Conclusion

A bandgap reference circuit was designed and simulated in Cadence Virtuoso. The core circuit was first optimized to achieve a low temperature coefficient. The optimized core used $R1 = 1 \text{ k}\Omega$, $R2 = 6.85 \text{ k}\Omega$, and BJT multipliers of 1:10:1. This produced a stable core voltage of about 1.17 V. Since the project required 1.4 V to 1.5 V, a non-inverting gain stage with gain 1.24 was added. The final output stayed around 1.45 V over the full temperature range from -40°C to 125°C . The estimated temperature coefficient was about 14.2 ppm/ $^\circ\text{C}$. Therefore, the final design satisfies both required specifications.